PHIVOLCS Special Report – The 22 April 2019 Ms 6.1 Central Luzon Earthquake

Epcenter: At 15.02°N, 120.34°E

PHIVOLCS Earthquake Intensity Scale (PEIS)	Affected Municipality/City
VIII (very destructive)	Brgy. Nabuclod, Floridablanca, and Brgy. Diaz, Porac, Pampanga
VII (destructive)	The rest of Floridablanca and Porac, Pampanga; Guagua, Lubao and Sasmuan, Pampanga; San Marcelino, and San Narciso, Zambales
VI (very strong)	Olongapo City, Subic, Zambales; Angeles City, Bacolor, & Masantol, Pampanga
V (strong)	Castillejos, Iba, and San Felipe, Zambales; Candaba, Macabebe, Magalang, Mexico, Minalin, and City of San Fernando, Pampanga; Abucay, Balanga, Mariveles, and Samal, Bataan; Malolos, Calumpit and Obando, Bulacan; Indang, Cavite; Sto. Domingo, Nueva Ecija; Lipa City; Makati City; Mandaluyong City; Manila City; Quezon City; Pasay City; San Juan City; Taguig City; Tarlac City; Valenzuela City
IV (moderately strong)	Meycauayan, Plaridel, and San Jose Del Monte City, Bulacan; San Rafael, Tarlac; Rosales and Villasis, Pangasinan; Itogon, and La Trinidad, Benguet; Kasibu, Nueva Vizcaya; Gabaldon, Nueva Ecija; San Mateo and Rodriguez, Rizal; Bacoor, Imus, and Maragondon, Cavite; Nasugbu, Batangas; Antipolo City; Baguio City; Caloocan City; Las Piñas City; Marikina City; Pasig City; Parañaque City; Tagaytay City
III (weak)	Marilao, Bulacan; Santo Domingo and Talavera, Nueva Ecija; Maddela, Quirino; Dingalan, Aurora; Infanta and Lucban, Quezon; Santa Cruz, and Calamba City, Laguna; Carmona, Dasmariñas, General Trias and Silang, Cavite; Lemery, San Nicolas and Talisay, Batangas; Cabanatuan City; Gapan City; Muntinlupa City; Palayan City
II (slightly felt)	Baler, Aurora; Dolores and Lucban, Quezon; Daet, Camarines Norte; Dagupan City, Lucena City
I (scarcely perceptible)	Sinait, Ilocos Sur; Calatagan, Batangas; Guinayangan, Quezon; Baguio City

Intensity VIII

Places	Coordinates	Findings (from the Method)
Nabuclod, Pampanga	15.0155° N, 120.4530° E	
Floridablanca, Pampanga	14.9619° N, 120.4947° E	
Diaz, Pampanga	15.0555° N, 120.5437° E	
Porac, Pampanga	15.0697° N, 120.5501° E	

Intensity-Based Epicenter Estimate Report

1) Initial (Unfiltered) Epicenter Estimate

Epicenter (H, K): 15.154399° N, 120.275078° E

Epicenter 1σ interval: (H ± 0.194209°, K ± 0.385870°)

Distance to PHIVOLCS Measured Epicenter: 16.490 km

Radii to (H, K)

Point	Radius (deg)	Radius (km)
P1	0.225719	24.565
P2	0.292044	31.848
P3	0.286249	30.863
P4	0.287769	30.989

Minimum R: 0.225719° (~24.565 km)

Maximum R: 0.292044° (~31.848 km)

Mean R_prox: 0.272945°

Radius std: 0.027349°

2) Recalculated Epicenter (After Outlier Removal)

Epicenter (H, K): 15.154399° N, 120.275078° E

Epicenter 1σ interval: (H ± 0.194209°, K ± 0.385870°)

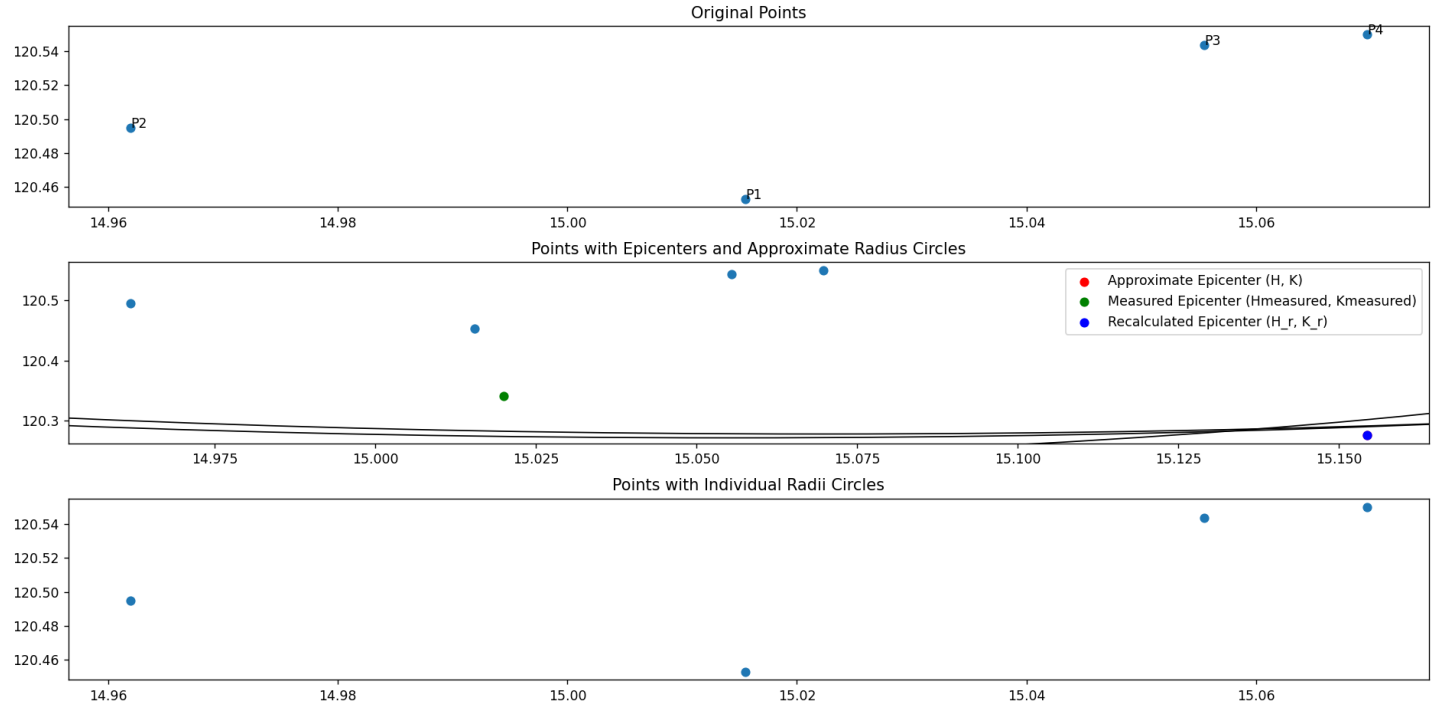
Distance to PHIVOLCS Measured Epicenter: 16.490 km

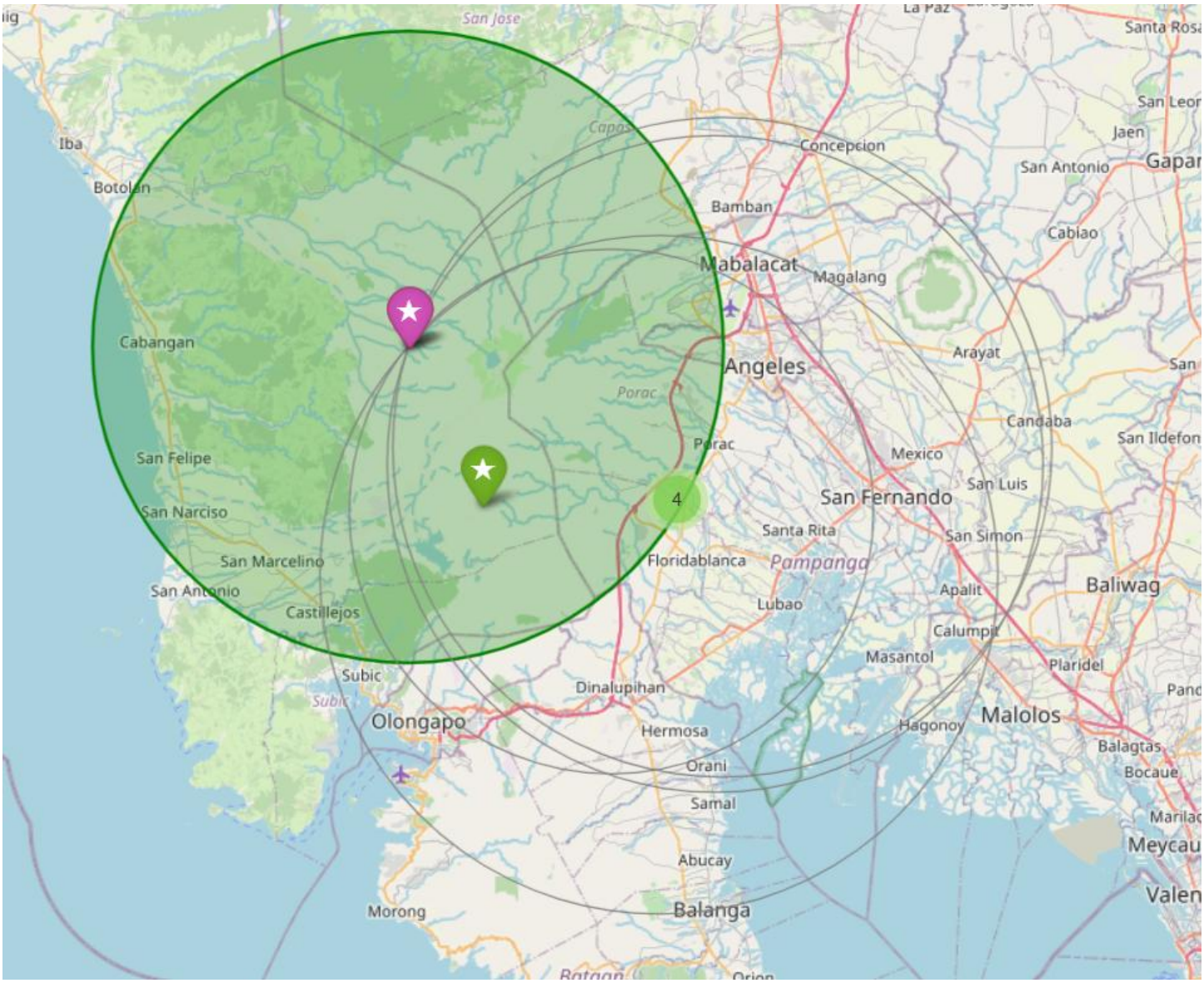
Radii to (H, K)

Point	Radius (deg)	Radius (km)
P1	0.225719	24.565
P2	0.292044	31.848
P3	0.286249	30.863
P4	0.287769	30.989

Minimum R: 0.225719° (~24.565 km)

Maximum R: 0.292044° (~31.848 km)


Original Points
P1
P2
P3
P4
120.54
120.52
120.50
120.48
120.46
14.96
14.98
15.00
15.02
15.04
15.06
Points with Epicenters and Approximate Radius Circles
Approximate Epicenter (H, K)
Measured Epicenter (Hmeasured, Kmeasured)
Recalculated Epicenter (H_r, K_r)
120.5
120.4
120.3
14.975
15.000
15.025
15.050
15.075
15.100
15.125
15.150
Points with Individual Radii Circles


San Jose
Iba
Botolan
Capas
Concepcion
Bamban
Mabalacat
Magalang
Cabangan
Angeles
Porac
San Felipe
San Narciso
San Marcelino
San Antonio
Castillejos
Subic
Olongapo
Floridablanca
Pampanga
San Fernando
Mexico
Santa Rita
Lubao
Dinalupihan
Hermosa
Orani
Samal
Abucay
Morong
Balanga
Malolos
Baliwag
Masantol
Calumpit
Apalit
San Simon
San Luis
Arayat
Candaba
Cabiao
San Antonio
Jaen
Santa Rosa
Hagonoy
Plaridel
Balagtas
Bocaue
4

Intensity VII

Places	Coordinates	Findings (from the Method)
Floridablanca, Pampanga	14.9619° N, 120.4947° E	
Porac, Pampanga	15.0697° N, 120.5501° E	
Guagua, Pampanga	14.9637° N, 120.6324° E	
Lubao, Pampanga	14.9268° N, 120.5879° E	
Sasmuan, Pampanga	14.8876° N, 120.6142° E	
San Marcelino, Zambales	15.0261° N, 120.2743° E	
San Narciso, Zambales	15.0111° N, 120.0896° E	

Intensity-Based Epicenter Estimate Report

1) Initial (Unfiltered) Epicenter Estimate

Epicenter (H, K): 14.929519° N, 120.388354° E

Epicenter 1σ interval: (H ± 0.797919°, K ± 0.196577°)

Distance to PHIVOLCS Measured Epicenter: 11.323 km

Radii to (H, K)

Point	Radius (deg)	Radius (km)
P1	0.111166	11.979
P2	0.214038	23.340
P3	0.246428	26.493
P4	0.199564	21.442
P5	0.229703	24.711
P6	0.149453	16.292
P7	0.309693	33.350

Minimum R: 0.111166° (~11.979 km)

Maximum R: 0.309693° (~33.350 km)

Mean R_prox: 0.208578°

Radius std: 0.059999°

2) Recalculated Epicenter (After Outlier Removal)

Epicenter (H, K): 14.894607° N, 120.420402° E

Epicenter 1σ interval: (H ± 0.504086°, K ± 0.150591°)

Distance to PHIVOLCS Measured Epicenter: 16.402 km

Radii to (H, K)

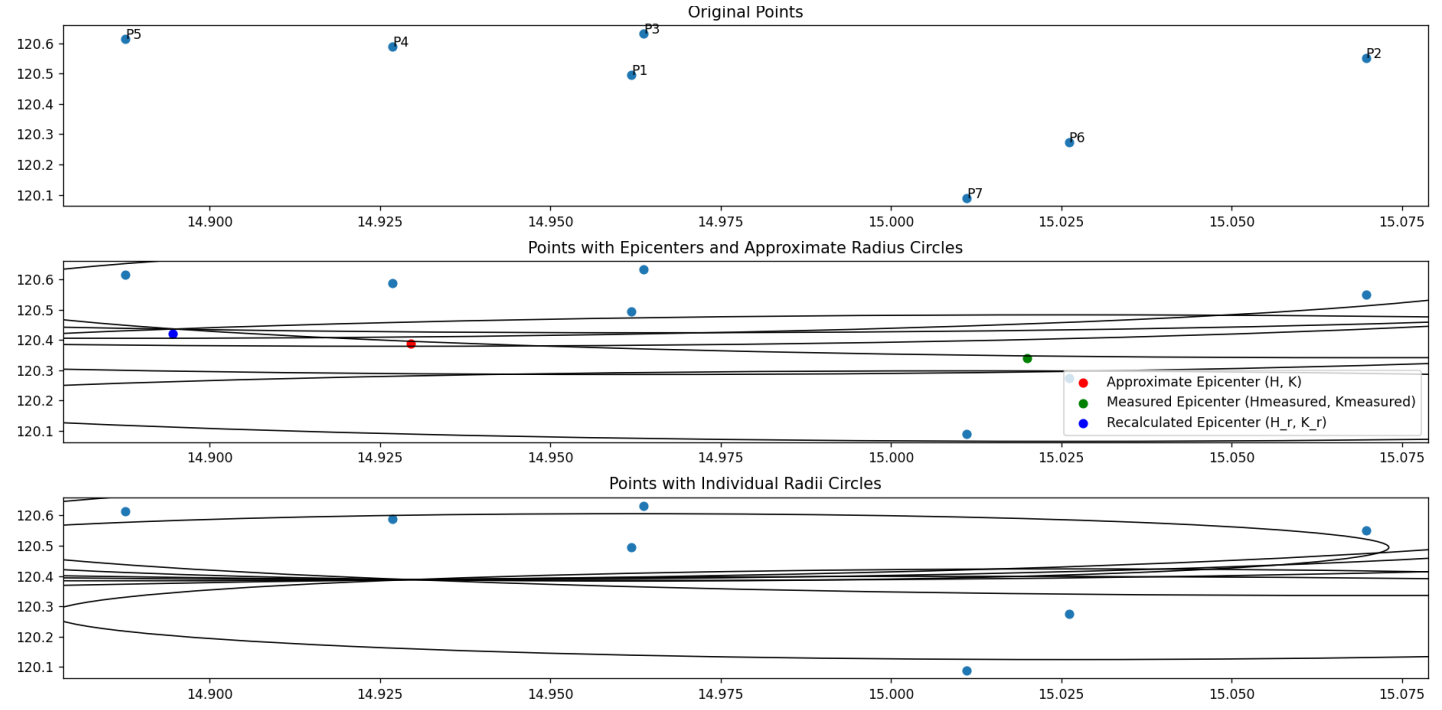
Point	Radius (deg)	Radius (km)
P3	0.222973	24.038
P5	0.193925	20.840

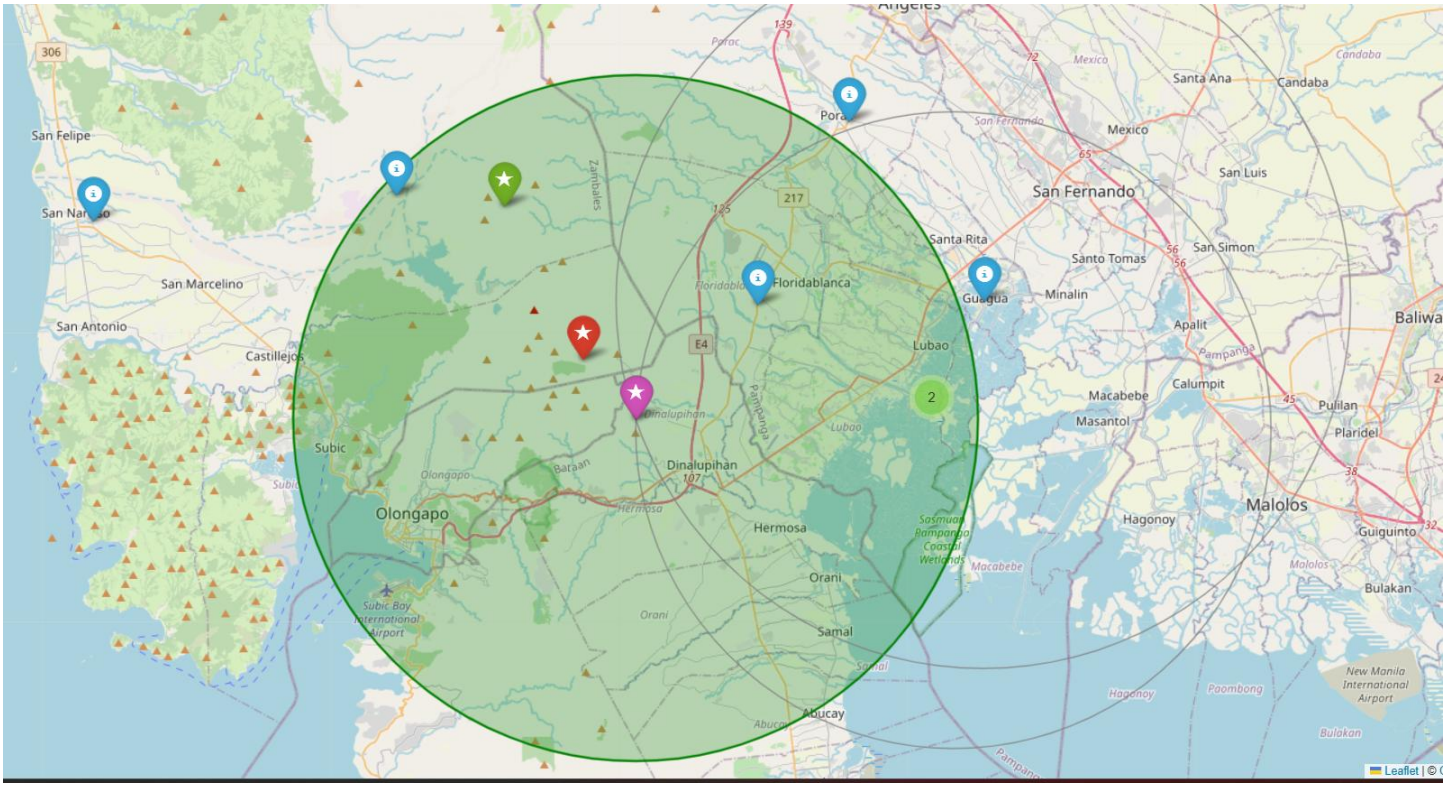
Minimum R: 0.193925° (~20.840 km)

Maximum R: 0.222973° (~24.038 km)

Mean R_prox: 0.208449°

Radius std: 0.014524°


Original Points
P5
P4
P3
P2
P1
P6
P7
Points with Epicenters and Approximate Radius Circles
Approximate Epicenter (H, K)
Measured Epicenter (Hmeasured, Kmeasured)
Recalculated Epicenter (H_r, K_r)
Points with Individual Radii Circles
120.6
120.5
120.4
120.3
120.2
120.1
14.900
14.925
14.950
14.975
15.000
15.025
15.050
15.075


San Felipe
San Marcelino
San Antonio
Castillejos
Subic
Olongapo
Dinalupihan
Hermosa
Orani
Samal
Abucay
Floridablanca
Santa Rita
Guagua
Lubao
San Fernando
Santa Ana
Candaba
Mexico
San Luis
Santo Tomas
San Simon
Minalin
Apalit
Macabebe
Masantol
Calumpit
Pulilan
Plaridel
Malolos
Hagonoy
Guiguinto
Bulakan
Baliwa
Subic Bay International Airport
New Manila International Airport
Zambales
Bataan
Pampanga
Leaflet

Intensity VI

Places	Coordinates	Findings (from the Method)
Olongapo City, Zambales	14.8436° N, 120.3090° E	
Subic, Zambales	14.8307° N, 120.1820° E	
Angeles City, Pampanga	15.1606° N, 120.6091° E	
Bacolor, Pampanga	14.9969° N, 120.6524° E	
Masantol, Pampanga	14.8567° N, 120.6833° E	

Intensity-Based Epicenter Estimate Report

1) Initial (Unfiltered) Epicenter Estimate

Epicenter (H, K): 14.578292° N, 120.065869° E

Epicenter 1σ interval: (H ± 1.223242°, K ± 1.029879°)

Distance to PHIVOLCS Measured Epicenter: 57.279 km

Radii to (H, K)

Point	Radius (deg)	Radius (km)
P1	0.359863	39.422
P2	0.277842	30.720
P3	0.796356	87.183
P4	0.720591	78.378
P5	0.677298	73.264

Minimum R: 0.277842° (~30.720 km)

Maximum R: 0.796356° (~87.183 km)

Mean R_prox: 0.566390°

Radius std: 0.207305°

2) Recalculated Epicenter (After Outlier Removal)

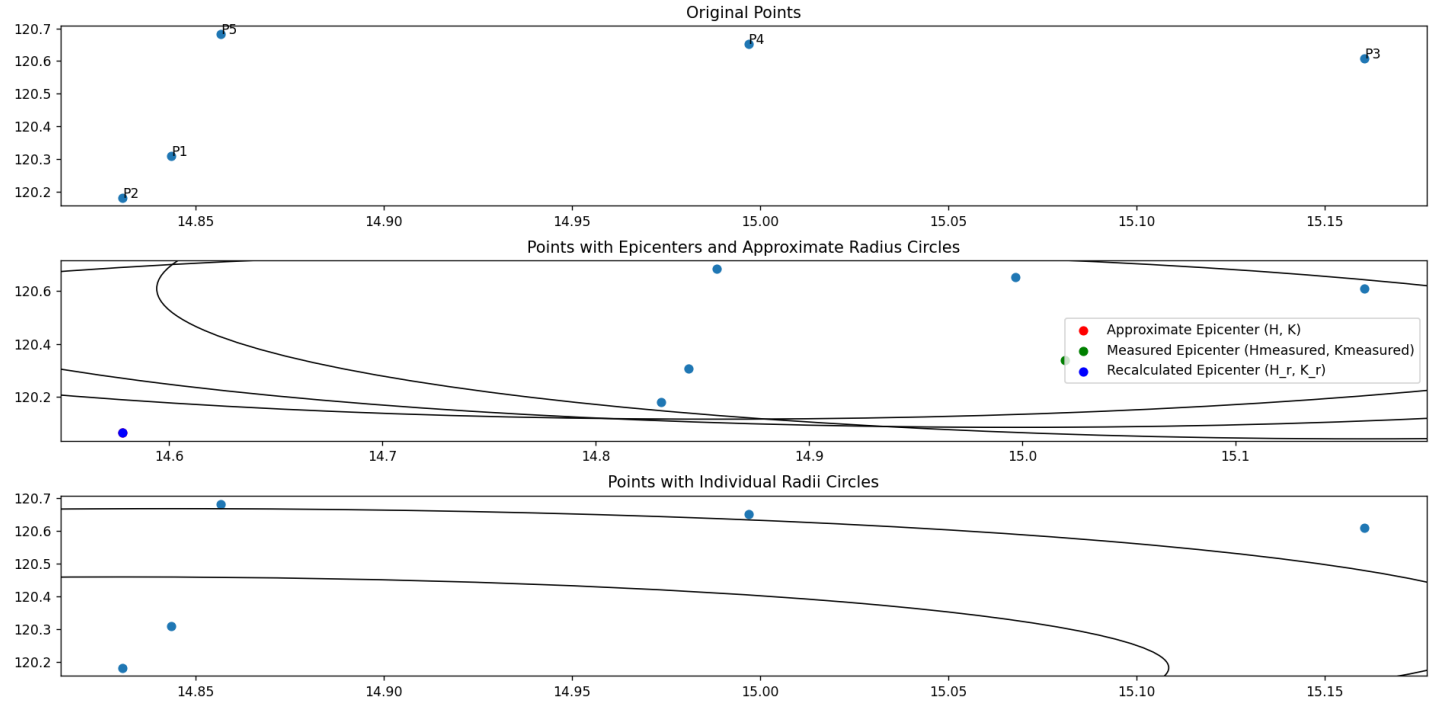
Epicenter (H, K): 14.578292° N, 120.065869° E

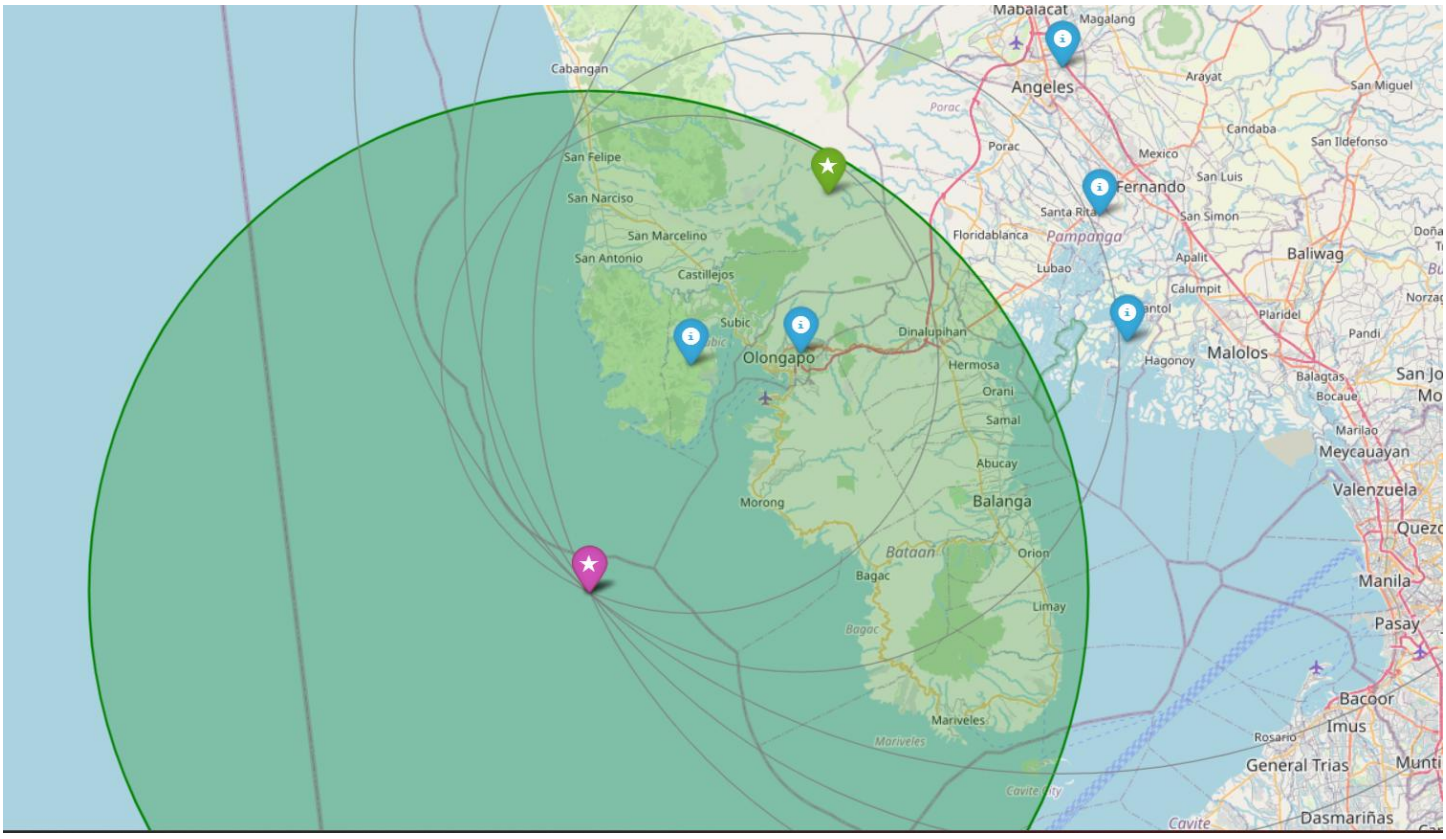
Epicenter 1σ interval: (H ± 1.223242°, K ± 1.029879°)

Distance to PHIVOLCS Measured Epicenter: 57.279 km

Radii to (H, K)

Point	Radius (deg)	Radius (km)
P1	0.359863	39.422
P2	0.277842	30.720
P3	0.796356	87.183
P4	0.720591	78.378
P5	0.677298	73.264


Original Points
P1
P2
P3
P4
P5
Points with Epicenters and Approximate Radius Circles
Approximate Epicenter (H, K)
Measured Epicenter (Hmeasured, Kmeasured)
Recalculated Epicenter (H_r, K_r)
Points with Individual Radii Circles


Mabalacat
Magalang
Angeles
Arayat
San Miguel
Cabangan
Porac
Candaba
San Ildefonso
Mexico
San Luis
San Felipe
San Narciso
Fernando
Santa Rita
San Simon
Floridablanca
Pampanga
Baliwag
San Marcelino
San Antonio
Lubao
Apalit
Castillejos
Calumpit
Plaridel
Pandi
Subic
Olongapo
Dinalupihan
Hermosa
Hagonoy
Malolos
Orani
Samal
Balagtas
Bocaue
Marilao
Meycauayan
Abucay
Valenzuela
Morong
Balanga
Bataan
Orion
Bagac
Manila
Limay
Pasay
Mariveles
Bacoor
Imus
Rosario
General Trias
Cavite City
Cavite
Dasmariñas

Intensity V

Places	Coordinates	Findings (from the Method)
Castillejos, Zambales	14.9438° N, 120.1936° E	
Iba, Zambales	15.3252° N, 119.9807° E	
San Felipe, Zambales	15.0793° N, 120.1243° E	
Candaba, Pampanga	15.0929° N, 120.8277° E	
Macabebe, Pampanga	14.9082° N, 120.7156° E	
Magalang, Pampanga	15.2139° N, 120.6618° E	
Mexico, Pampanga	15.0646° N, 120.7203° E	
Minalin, Pampanga	14.9705° N, 120.6852° E	
San Fernando City, Pampanga	15.0278° N, 120.6935° E	
Abucay, Bataan	14.7220° N, 120.5354° E	
Balanga, Bataan	14.6799° N, 120.5421° E	
Mariveles, Bataan	14.4354° N, 120.4900° E	
Samal, Bataan	14.7690° N, 120.5428° E	
Malolos, Bulacan	14.8605° N, 120.8092° E	
Calumpit, Bulacan	14.9152° N, 120.7662° E	
Obando, Bulacan	14.7099° N, 120.9367° E	
Indang, Cavite	14.1920° N, 120.8729° E	
Sto. Domingo, Nueva Ecija	15.5922° N, 120.8788° E	
Lipa City	13.9411° N, 121.1624° E	
Makati City	14.5574° N, 121.0227° E	
Mandaluyong City	14.5763° N, 121.0392° E	
Manila City	14.5995° N, 120.9842° E	
Quezon City	14.6488° N, 121.0509° E	
Pasay City	14.5377° N, 121.0008° E	
San Juan City	14.6020° N, 121.0355° E	
Taguig City	14.5306° N, 121.0575° E	
Tarlac City	15.4881° N, 120.5872° E	
Valenzuela City	14.6925° N, 120.9699° E	

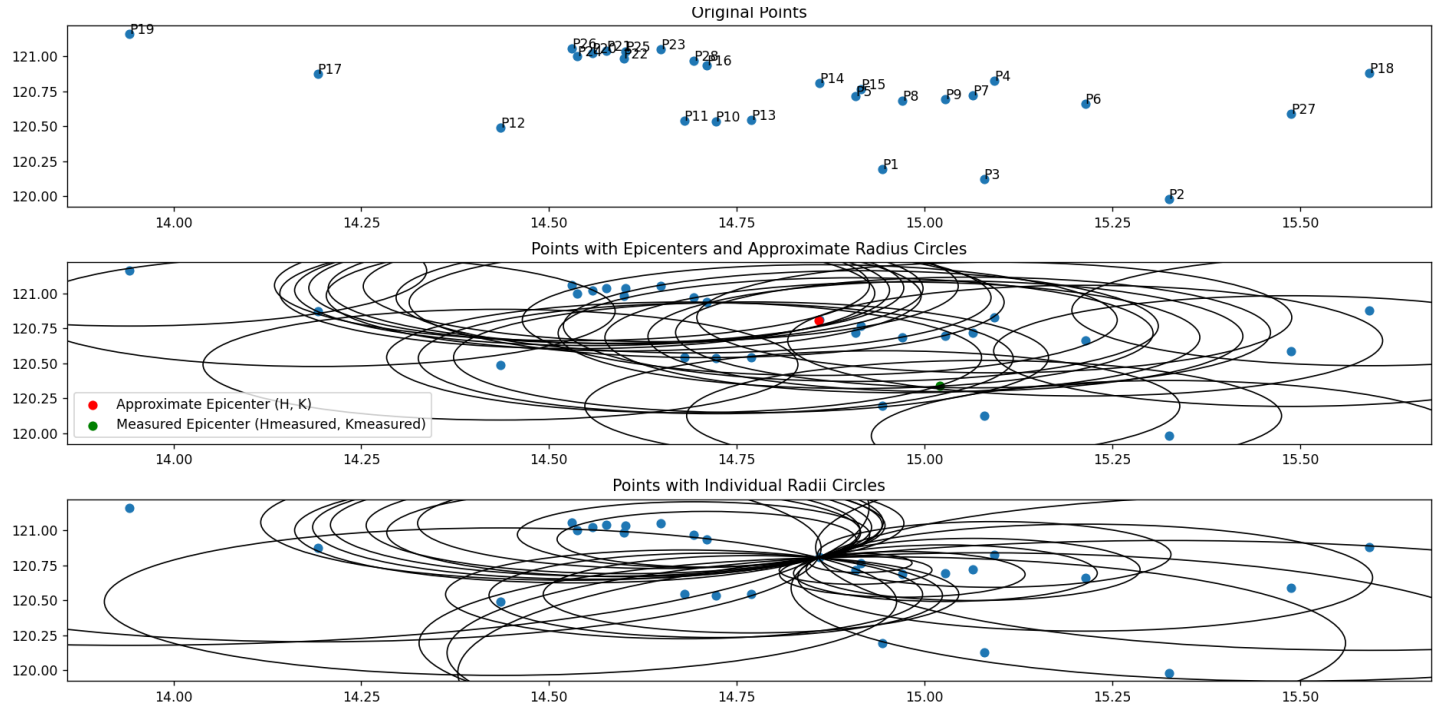
Intensity-Based Epicenter Estimate Report

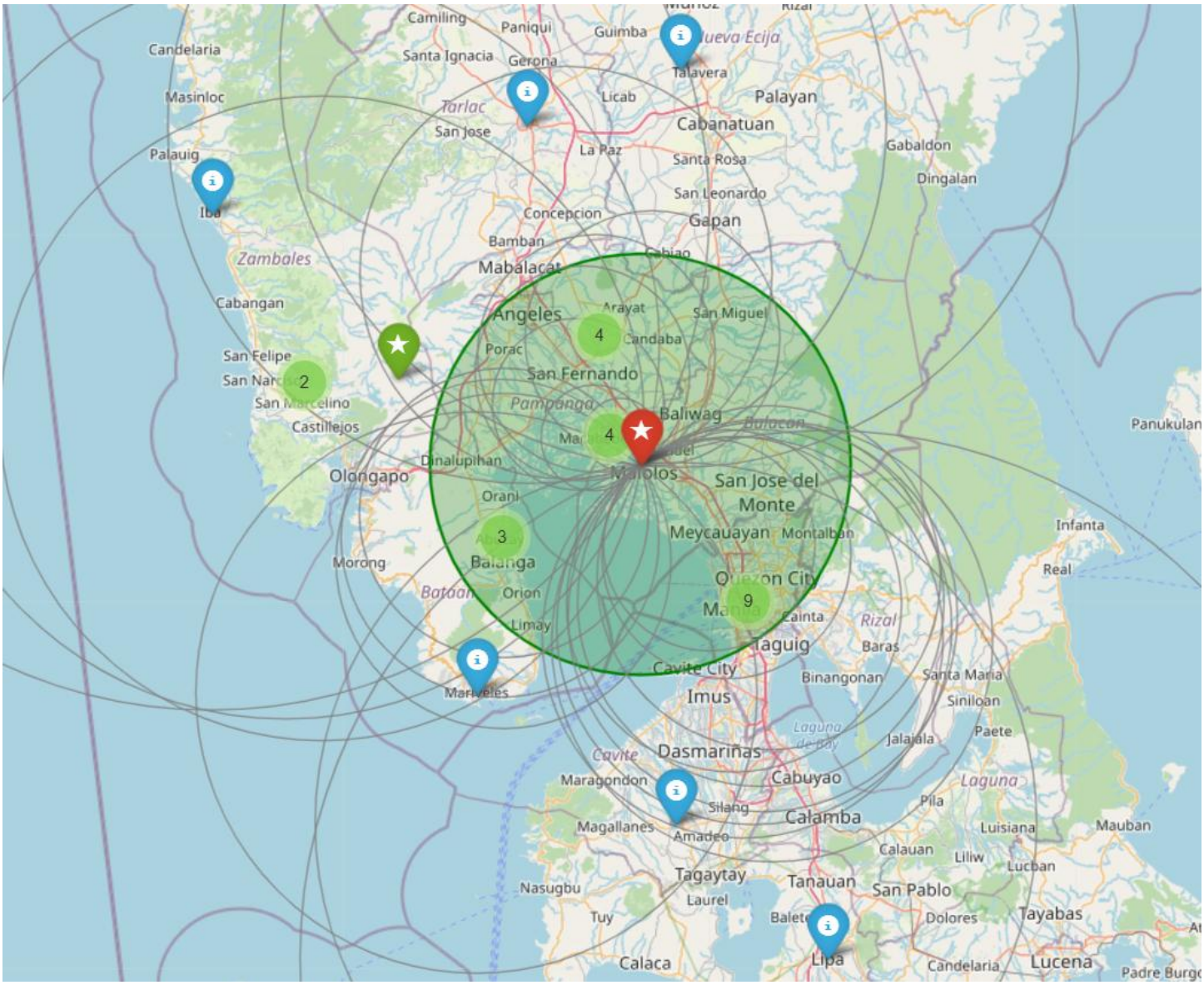
1) Initial (Unfiltered) Epicenter Estimate

Epicenter (H, K): 14.858805° N, 120.804496° E

Epicenter 1σ interval: (H ± 4.953289°, K ± 12.007391°)

Distance to PHIVOLCS Measured Epicenter: 53.025 km


Original Points
P19
P17
P26
P24
P21
P22
P25
P23
P28
P16
P14
P15
P5
P8
P9
P7
P4
P18
P6
P27
P12
P11
P10
P13
P1
P3
P2
Points with Epicenters and Approximate Radius Circles
Approximate Epicenter (H, K)
Measured Epicenter (Hmeasured, Kmeasured)
Points with Individual Radii Circles
14.00
14.25
14.50
14.75
15.00
15.25
15.50
120.00
120.25
120.50
120.75
121.00


Camiling
Paniqui
Guimba
Talavera
Candelaria
Santa Ignacia
Gerona
Masinloc
Tarlac
Licab
Palayan
San Jose
Cabanatuan
La Paz
Palauig
Santa Rosa
Gabaldon
Dingalan
San Leonardo
Iba
Concepcion
Gapan
Bamban
Zambales
Mabalacat
Cabangan
Angeles
Arayat
San Miguel
Candaba
Porac
San Felipe
San Fernando
Pampanga
Castillejos
Baliwag
Olongapo
Dinalupihan
Orani
San Jose del Monte
Meycauayan
Montalban
Infanta
Morong
Balanga
Bataan
Orion
Real
Limay
Rizal
Cainta
Taguig
Baras
Cavite City
Binangonan
Santa Maria
Imus
Siniloan
Laguna de Bay
Jalajala
Paete
Cavite
Dasmariñas
Maragondon
Cabuyao
Laguna
Pila
Silang
Calamba
Magallanes
Amadeo
Luisiana
Mauban
Calauan
Liliw
Lucban
Tagaytay
Tanauan
San Pablo
Nasugbu
Laurel
Tuy
Dolores
Tayabas
Calaca
Lipa
Candelaria
Lucena
Padre Burgos
Panukulan

Intensity IV

Places	Coordinates	Findings (from the Method)
Meycauayan, Bulacan	14.7377° N, 120.9608° E	
Plaridel, Bulacan	14.8859° N, 120.8593° E	
San Jose Del Monte City, Bulacan	14.8287° N, 121.0432° E	
San Rafael, Tarlac	15.4572° N, 120.5883° E	
Rosales, Pangasinan	15.8613° N, 120.6315° E	
Villasis, Pangasinan	15.9052° N, 120.5509° E	
Itogon, Benguet	16.3612° N, 120.6787° E	
La Trinidad, Benguet	16.4484° N, 120.5905° E	
Kasibu, Nueva Vizcaya	16.3666° N, 121.3313° E	
Gabaldon, Nueva Ecija	15.5155° N, 121.3084° E	
San Mateo, Rizal	14.6954° N, 121.1177° E	
Rodriguez, Rizal	14.7320° N, 121.1452° E	
Bacoor, Cavite	14.4312° N, 120.9657° E	
Imus, Cavite	14.4276° N, 120.9372° E	
Maragondon, Cavite	14.2761° N, 120.7374° E	
Nasugbu, Batangas	14.0940° N, 120.6890° E	
Antipolo City	14.5847° N, 121.1757° E	
Baguio City	16.4130° N, 120.5914° E	
Caloocan City	14.6514° N, 120.9902° E	
Las Pinas City	14.4493° N, 120.9819° E	
Marikina City	14.6349° N, 121.1020° E	
Pasig City	14.5608° N, 121.0776° E	
Paranaque City	14.4793° N, 121.0198° E	
Tagaytay City	14.1154° N, 120.9620° E	

Intensity-Based Epicenter Estimate Report

1) Initial (Unfiltered) Epicenter Estimate

Epicenter (H, K): 15.239858° N, 121.096333° E
Epicenter 1σ interval: (H ± 9.101865°, K ± 40.699402°)
Distance to PHIVOLCS Measured Epicenter: 84.786 km

Radii to (H, K)

Point	Radius (deg)	Radius (km)
P1	0.520127	57.704
P2	0.425994	46.871
P3	0.414577	46.073
P4	0.552571	59.596
P5	0.776053	85.173
P6	0.860335	94.269
P7	1.196588	132.452
P8	1.310129	144.868
P9	1.150981	127.785
P10	0.347780	38.162
P11	0.544878	60.585
P12	0.510204	56.715
P13	0.819142	91.009
P14	0.827700	91.925
P15	1.028427	113.903
P16	1.216105	134.737
P17	0.659948	73.348
P18	1.277191	141.189
P19	0.597953	66.420
P20	0.798798	88.762
P21	0.604985	67.271
P22	0.679317	75.535
P23	0.764399	84.969
P24	1.132454	125.866

Minimum R: 0.347780° (~38.162 km)
Maximum R: 1.310129° (~144.868 km)
Mean R_prox: 0.792360°
Radius std: 0.288968°

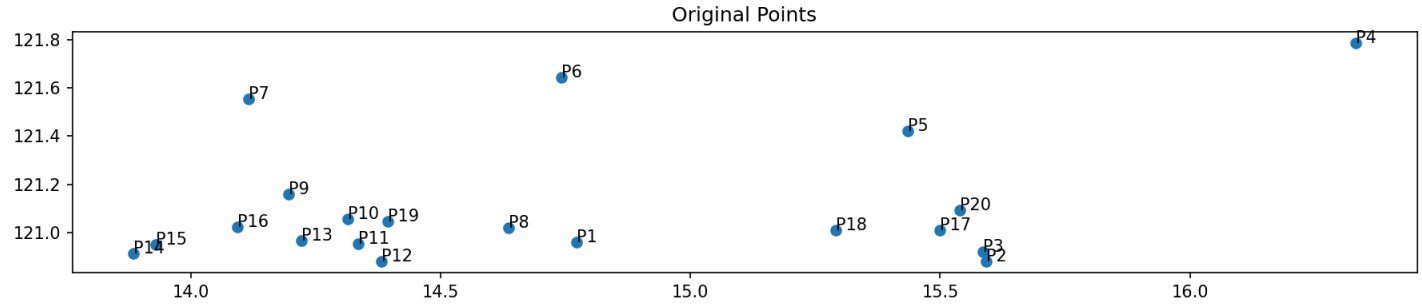
2) Recalculated Epicenter (After Outlier Removal)

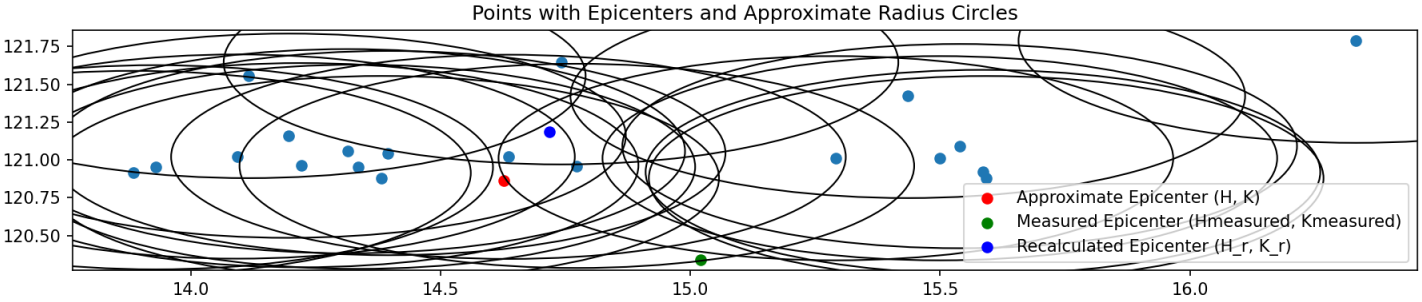
Epicenter (H, K): 15.130520° N, 120.785544° E
Epicenter 1σ interval: (H ± 1.572784°, K ± 6.495611°)
Distance to PHIVOLCS Measured Epicenter: 49.391 km

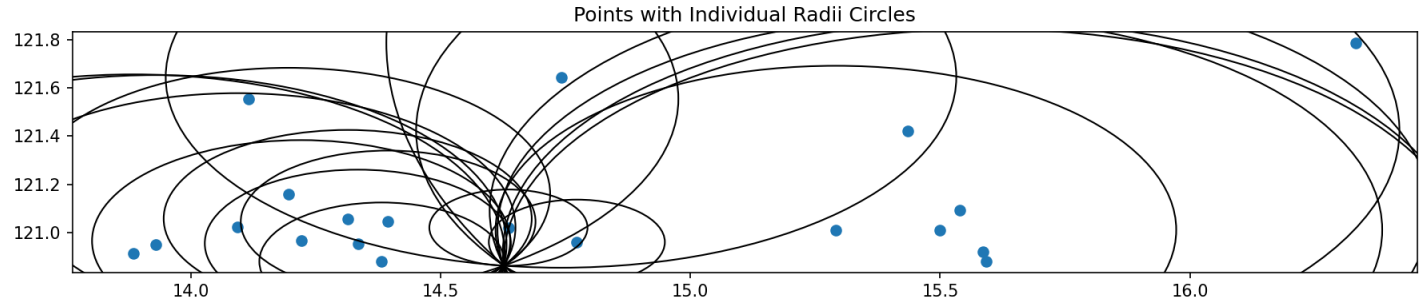
Radii to (H, K)

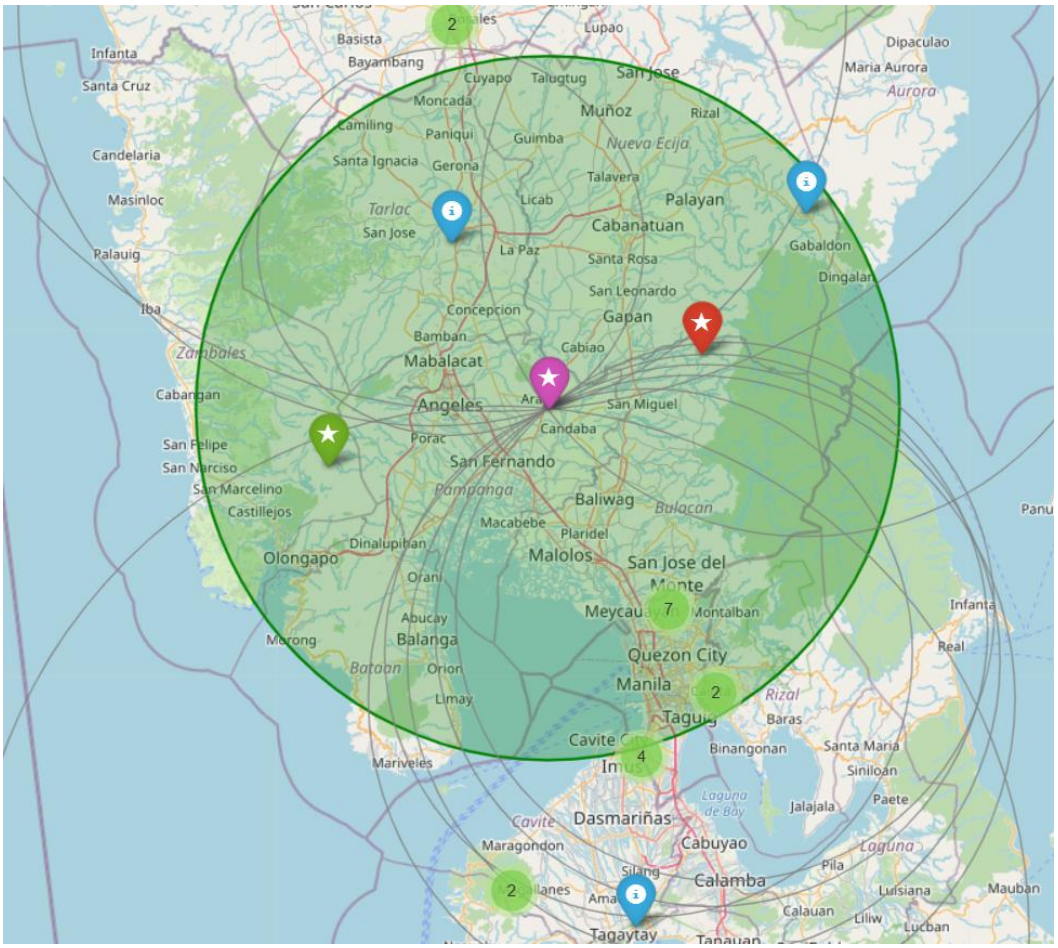
Point	Radius (deg)	Radius (km)
P4	0.381609	42.037
P6	0.809436	89.734
P7	1.235310	137.322
P10	0.649298	70.545
P11	0.547409	60.122
P12	0.536815	58.795
P16	1.041006	115.723
P19	0.520999	57.637
P21	0.588033	64.759
P22	0.640217	70.701

Minimum R: 0.381609° (~42.037 km)
Maximum R: 1.235310° (~137.322 km)
Mean R_prox: 0.695013°
Radius std: 0.248310°


Original Points
P1
P2
P3
P4
P5
P6
P7
P8
P9
P10
P11
P12
P13
P14
P15
P16
P17
P18
P19
P20
121.8
121.6
121.4
121.2
121.0
14.0
14.5
15.0
15.5
16.0


Points with Epicenters and Approximate Radius Circles
121.75
121.50
121.25
121.00
120.75
120.50
Approximate Epicenter (H, K)
Measured Epicenter (Hmeasured, Kmeasured)
Recalculated Epicenter (H_r, K_r)
14.0
14.5
15.0
15.5
16.0


Points with Individual Radii Circles
121.8
121.6
121.4
121.2
121.0
14.0
14.5
15.0
15.5
16.0


Basista
Bayambang
Lupao
Dipaculao
Infanta
Santa Cruz
Cuyapo
Talugtug
San Jose
Maria Aurora
Aurora
Moncada
Camiling
Paniqui
Guimba
Muñoz
Rizal
Nueva Ecija
Candelaria
Santa Ignacia
Gerona
Talavera
Masinloc
Tarlac
Licab
Palayan
San Jose
Cabanatuan
La Paz
Gabaldon
Palauig
Santa Rosa
Dingalan
San Leonardo
Iba
Concepcion
Gapan
Bamban
Cabiao
Zambales
Mabalacat
Cabangan
Angeles
San Miguel
Candaba
San Felipe
Porac
San Fernando
San Narciso
San Marcelino
Pampanga
Castillejos
Baliwag
Bulacan
Macabebe
Plaridel
Dinalupihan
Olongapo
Malolos
San Jose del
Orani
Meycau
Montalban
Abucay
Balanga
Morong
Quezon City
Bataan
Orion
Manila
Limay
Rizal
Taguig
Baras
Cavite
Binangonan
Santa Maria
Mariveles
Siniloan
Laguna de Bay
Jalajala
Paete
Cavite
Dasmariñas
Maragondon
Cabuyao
Laguna
Pila
Silang
Calamba
Luisiana
Mauban
Calauan
Liliw
Lucban
Tagaytay

Intensity III

Places	Coordinates	Findings (from the Method)
Marilao, Bulacan	14.7724° N, 120.9600° E	
Santo Domingo, Nueva Ecija	15.5922° N, 120.8788° E	
Talavera, Nueva Ecija	15.5857° N, 120.9197° E	
Maddela, Quirino	16.3326° N, 121.7879° E	
Dingalan, Aurora	15.4352° N, 121.4228° E	
Infanta, Quezon	14.7407° N, 121.6451° E	
Lucban, Quezon	14.1144° N, 121.5547° E	
Santa Cruz, Laguna	14.6354° N, 121.0199° E	
Calamba City, Laguna	14.1949° N, 121.1597° E	
Carmona, Cavite	14.3135° N, 121.0571° E	
Dasmarinas, Cavite	14.3338° N, 120.9542° E	
General Trias, Cavite	14.3816° N, 120.8792° E	
Silang, Cavite	14.2202° N, 120.9647° E	
Lemery, Batangas	13.8845° N, 120.9131° E	
San Nicholas, Batangas	13.9289° N, 120.9508° E	
Talisay, Batangas	14.0928° N, 121.0217° E	
Cabanatuan City	15.4994° N, 121.0106° E	
Gapan City	15.2912° N, 121.0106° E	
Muntilupa City	14.3945° N, 121.0448° E	
Palayan City	15.5391° N, 121.0908° E	

Intensity-Based Epicenter Estimate Report

1) Initial (Unfiltered) Epicenter Estimate

Epicenter (H, K): 14.626037° N, 120.861780° E
Epicenter 1σ interval: (H ± 3.716257°, K ± 13.336585°)
Distance to PHIVOLCS Measured Epicenter: 71.168 km

Radii to (H, K)

Point	Radius (deg)	Radius (km)
P1	0.176265	19.403
P2	0.966313	107.448
P3	0.961409	106.891
P4	1.941663	214.145
P5	0.984626	108.283
P6	0.791667	85.216
P7	0.861342	93.848
P8	0.158397	17.044
P9	0.524056	57.686
P10	0.368550	40.620
P11	0.306502	33.985
P12	0.245057	27.245
P13	0.418684	46.468
P14	0.743310	82.641
P15	0.702797	78.109
P16	0.556701	61.745
P17	0.885952	98.420
P18	0.681608	75.671
P19	0.295136	32.419
P20	0.941347	104.463

Minimum R: 0.158397° (~17.044 km)
Maximum R: 1.941663° (~214.145 km)
Mean R_prox: 0.675569°
Radius std: 0.400836°

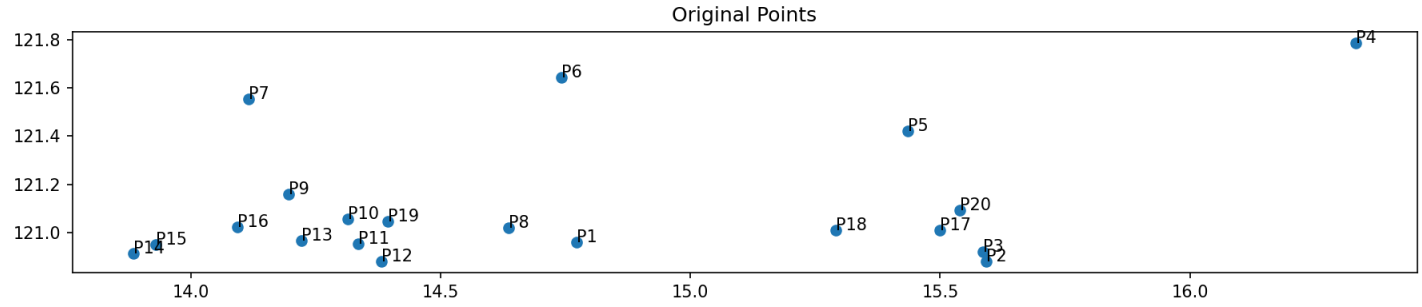
2) Recalculated Epicenter (After Outlier Removal)

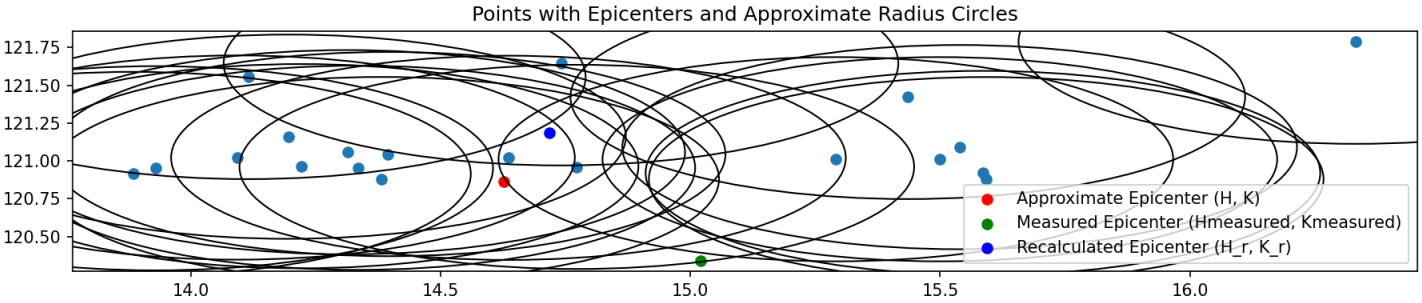
Epicenter (H, K): 14.717050° N, 121.185229° E
Epicenter 1σ interval: (H ± 0.965530°, K ± 4.024283°)
Distance to PHIVOLCS Measured Epicenter: 96.883 km

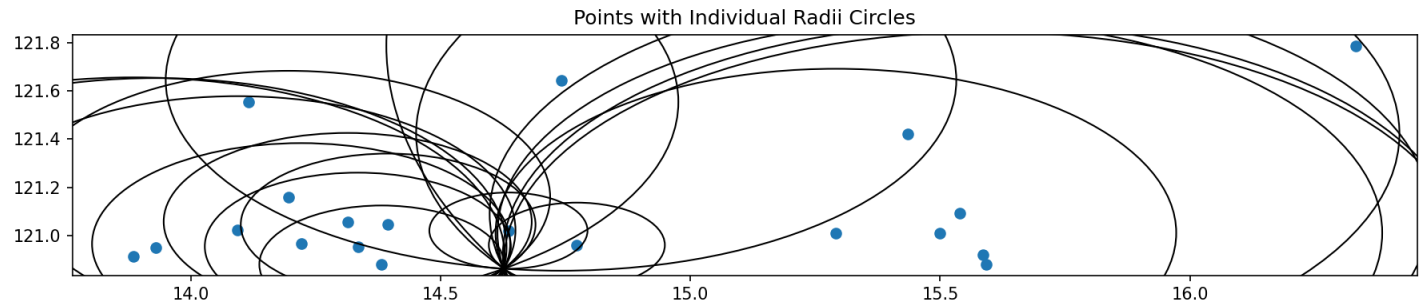
Radii to (H, K)

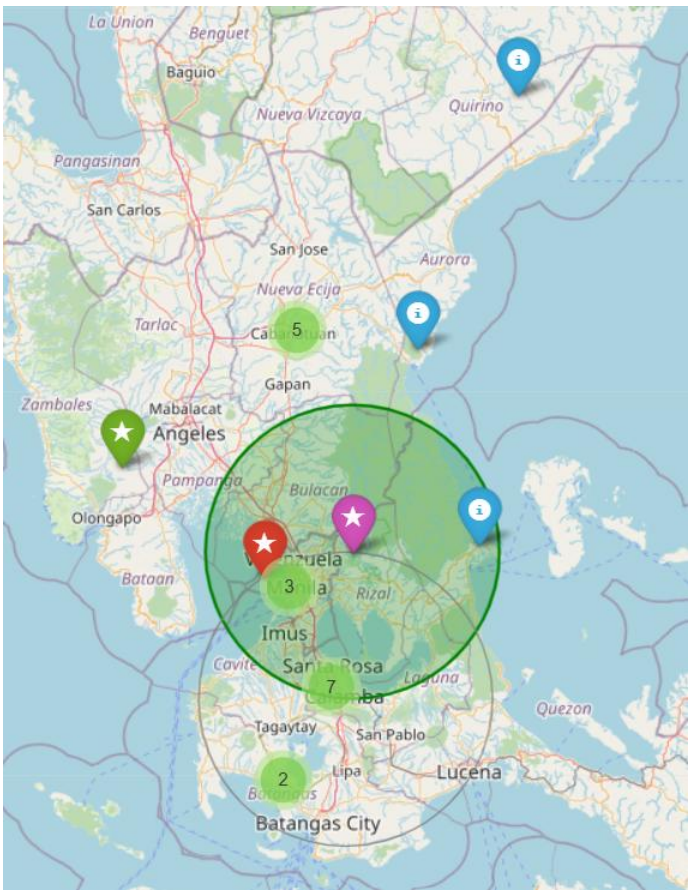
Point	Radius (deg)	Radius (km)
P9	0.522774	58.126

Minimum R: 0.522774° (~58.126 km)
Maximum R: 0.522774° (~58.126 km)
Mean R_prox: 0.522774°
Radius std: 0.000000°


Original Points
P1
P2
P3
P4
P5
P6
P7
P8
P9
P10
P11
P12
P13
P14
P15
P16
P17
P18
P19
P20
121.8
121.6
121.4
121.2
121.0
14.0
14.5
15.0
15.5
16.0


Points with Epicenters and Approximate Radius Circles
Approximate Epicenter (H, K)
Measured Epicenter (Hmeasured, Kmeasured)
Recalculated Epicenter (H_r, K_r)
121.75
121.50
121.25
121.00
120.75
120.50
14.0
14.5
15.0
15.5
16.0


Points with Individual Radii Circles
121.8
121.6
121.4
121.2
121.0
14.0
14.5
15.0
15.5
16.0


La Union
Benguet
Baguio
Nueva Vizcaya
Quirino
Pangasinan
San Carlos
San Jose
Aurora
Nueva Ecija
Tarlac
Gapan
Zambales
Mabalacat
Angeles
Pampanga
Bulacan
Olongapo
Bataan
Rizal
Imus
Cavite
Laguna
Quezon
Tagaytay
San Pablo
Lipa
Lucena
Batangas City
5
3
7
2

Intensity II

Places	Coordinates	Findings (from the Method)
Baler, Aurora	15.5391° N, 121.0908° E	
Dolores, Quezon	14.0135° N, 121.3909° E	
Lucban, Quezon	14.1144° N, 121.5547° E	
Daet, Camarines Norte	14.1030° N, 122.9325° E	
Dagupan City	16.0337° N, 120.3320° E	
Lucena City	13.9414° N, 121.6235° E	

Intensity-Based Epicenter Estimate Report

1) Initial (Unfiltered) Epicenter Estimate

Epicenter (H, K): 14.137405° N, 121.281832° E

Epicenter 1σ interval: (H ± 3.572096°, K ± 4.054152°)

Distance to PHIVOLCS Measured Epicenter: 141.082 km

Radii to (H, K)

Point	Radius (deg)	Radius (km)
P1	1.414653	157.208
P2	0.165070	18.116
P3	0.273836	29.535
P4	1.651027	178.041
P5	2.120876	234.221
P6	0.393897	42.819

Minimum R: 0.165070° (~18.116 km)

Maximum R: 2.120876° (~234.221 km)

Mean R_prox: 1.003227°

Radius std: 0.757612°

2) Recalculated Epicenter (After Outlier Removal)

Epicenter (H, K): 14.671823° N, 121.194888° E

Epicenter 1σ interval: (H ± 1.053682°, K ± 1.638695°)

Distance to PHIVOLCS Measured Epicenter: 99.709 km

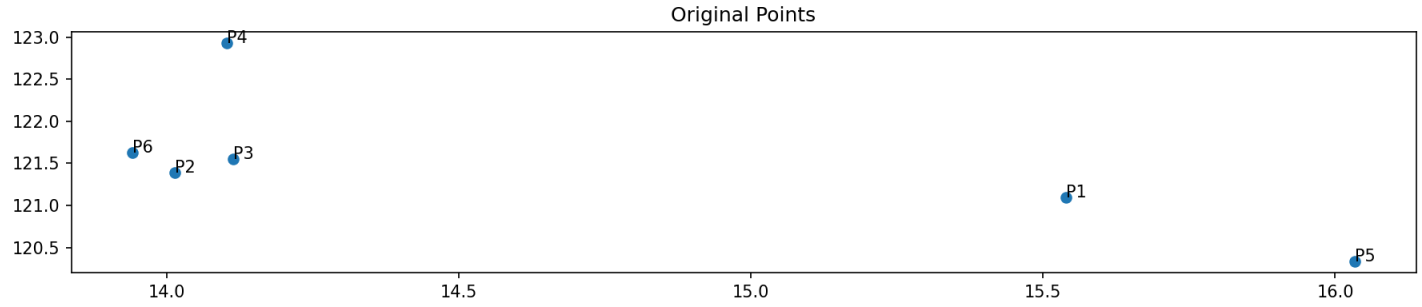
Radii to (H, K)

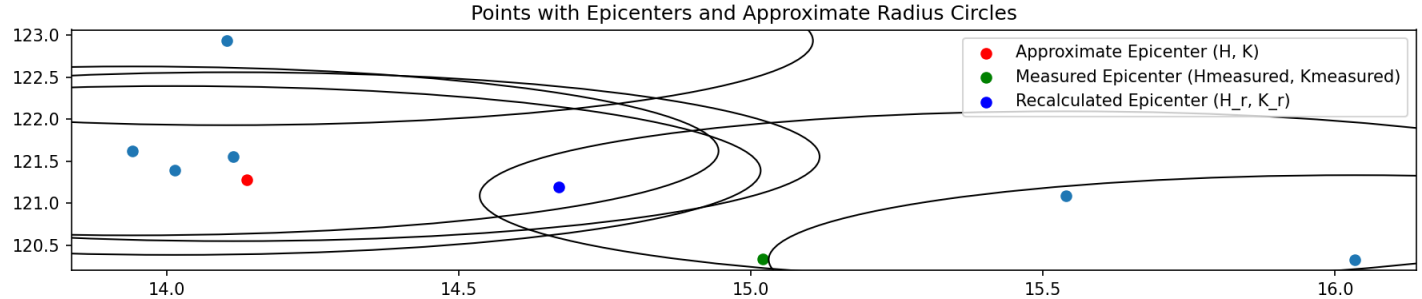
Point	Radius (deg)	Radius (km)
P2	0.686884	76.187

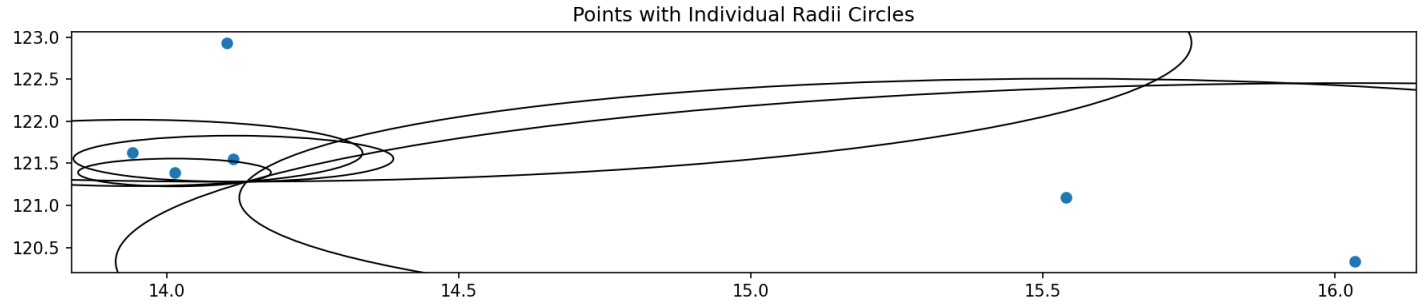
Minimum R: 0.686884° (~76.187 km)

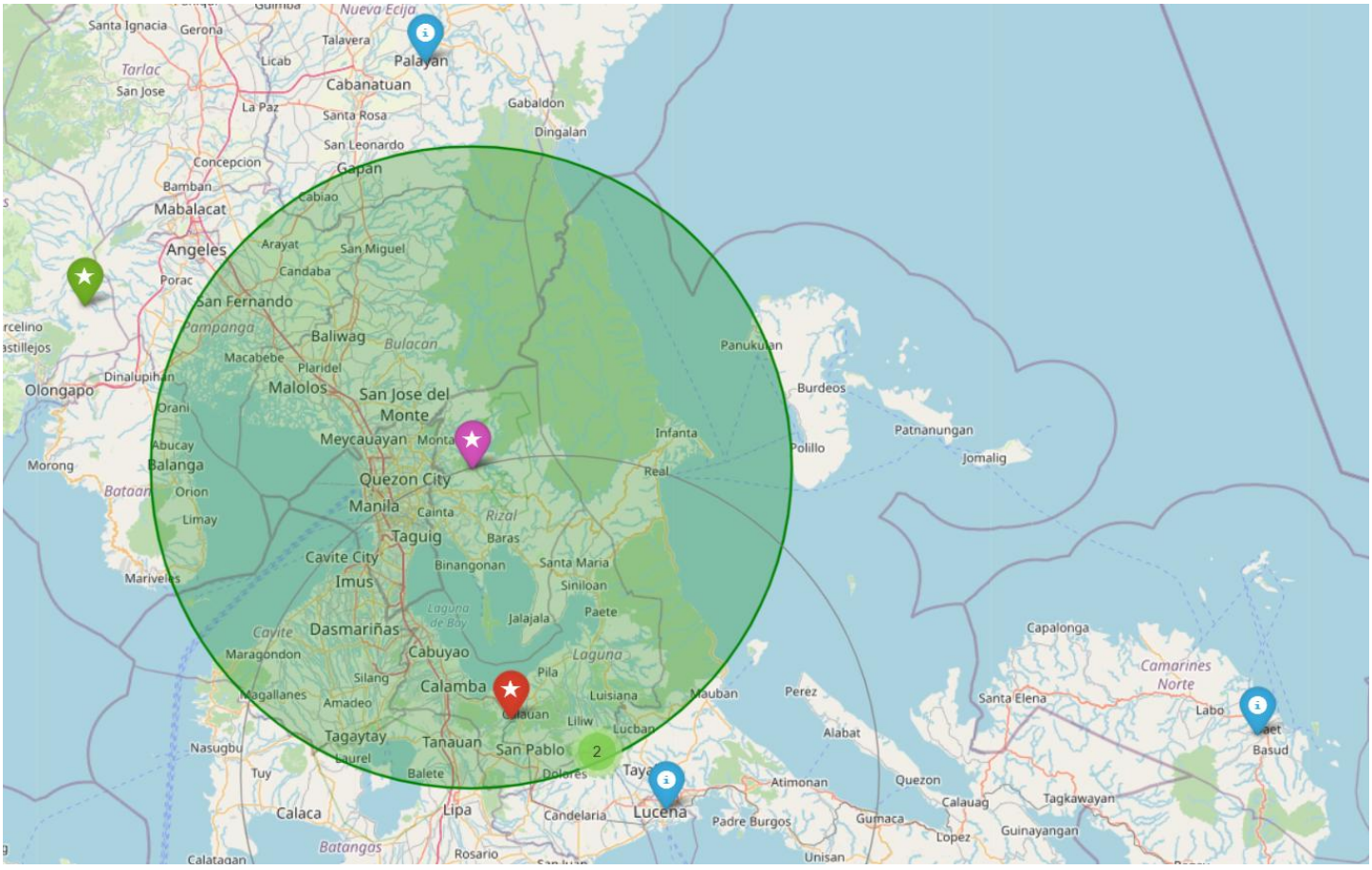
Maximum R: 0.686884° (~76.187 km)

Mean R_prox: 0.686884°


Original Points
P4
P6
P2
P3
P1
P5
123.0
122.5
122.0
121.5
121.0
120.5
14.0
14.5
15.0
15.5
16.0


Points with Epicenters and Approximate Radius Circles
Approximate Epicenter (H, K)
Measured Epicenter (Hmeasured, Kmeasured)
Recalculated Epicenter (H_r, K_r)
123.0
122.5
122.0
121.5
121.0
120.5
14.0
14.5
15.0
15.5
16.0


Points with Individual Radii Circles
123.0
122.5
122.0
121.5
121.0
120.5
14.0
14.5
15.0
15.5
16.0


Nueva Ecija
Santa Ignacia
Gerona
Talavera
Palayan
Tarlac
Licab
San Jose
Cabanatuan
La Paz
Santa Rosa
Gabaldon
Dingalan
San Leonardo
Concepcion
Gapan
Bamban
Cabiao
Mabalacat
Angeles
Arayat
San Miguel
Candaba
Porac
San Fernando
Pampanga
Baliwag
Bulacan
Macabebe
Plaridel
Dinalupihan
Olongapo
Malolos
San Jose del Monte
Orani
Meycauayan
Abucay
Balanga
Morong
Bataan
Orion
Limay
Quezon City
Manila
Cainta
Rizal
Taguig
Baras
Cavite City
Binangonan
Santa Maria
Mariveles
Imus
Siniloan
Laguna de Bay
Jalajala
Paete
Cavite
Dasmariñas
Maragondon
Cabuyao
Laguna
Silang
Pila
Calamba
Magallanes
Amadeo
Luisiana
Mauban
Perez
Capalonga
Camarines Norte
Santa Elena
Labo
Basud
Alabat
Tagaytay
Tanauan
San Pablo
Nasugbu
Laurel
Balete
Dolores
Tuy
Atimonan
Quezon
Calauag
Tagkawayan
Calaca
Lipa
Candelaria
Lucena
Padre Burgos
Gumaca
Lopez
Guinayangan
Batangas
Rosario
Unisan
Panukulan
Burdeos
Patnanungan
Jomalig
Infanta
Polillo
Real

Intensity I

Places	Coordinates	Findings (from the Method)
Sinait, Ilocos Sur	17.8535° N, 120.5049° E	
Calatagan, Batangas	13.8820° N, 120.6545° E	
Guinayangan, Quezon	13.8851° N, 122.4241° E	
Baguio City	16.4130° N, 120.5914° E	

Intensity-Based Epicenter Estimate Report

1) Initial (Unfiltered) Epicenter Estimate

Epicenter (H, K): 15.545975° N, 107.900197° E

Epicenter 1σ interval: (H ± 1.313068°, K ± 25.405292°)

Distance to PHIVOLCS Measured Epicenter: 1,335.420 km

Radii to (H, K)

Point	Radius (deg)	Radius (km)
P1	12.814180	1,366.434
P2	12.862389	1,383.889
P3	14.618559	1,572.558
P4	12.720785	1,359.866

Minimum R: 12.720785° (~1,359.866 km)

Maximum R: 14.618559° (~1,572.558 km)

Mean R_prox: 13.253978°

Radius std: 0.789484°

2) Recalculated Epicenter (After Outlier Removal)

Epicenter (H, K): 15.545975° N, 107.900197° E

Epicenter 1σ interval: (H ± 1.313068°, K ± 25.405292°)

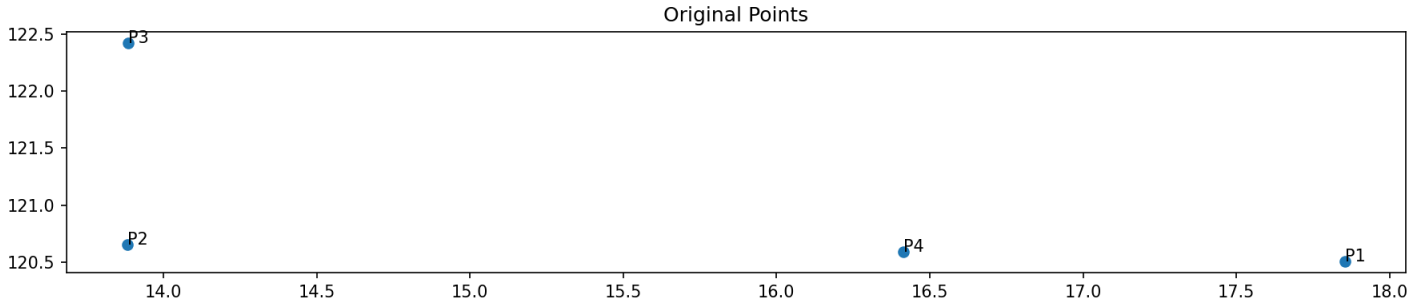
Distance to PHIVOLCS Measured Epicenter: 1,335.420 km

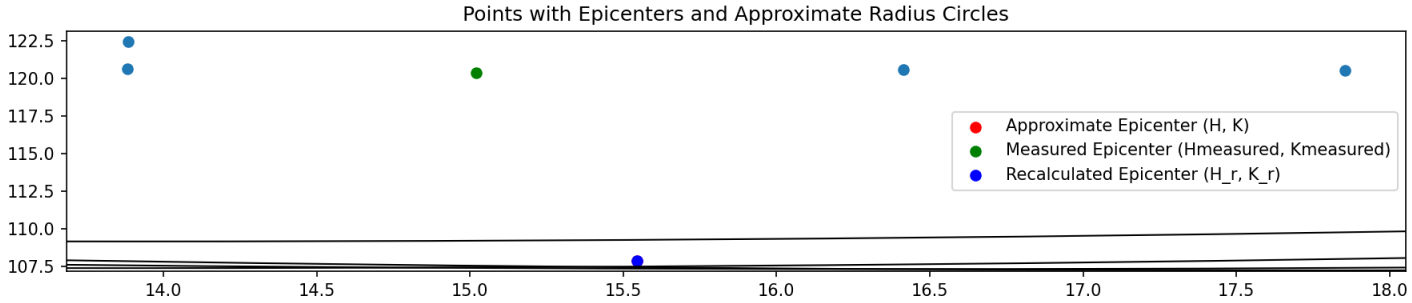
Radii to (H, K)

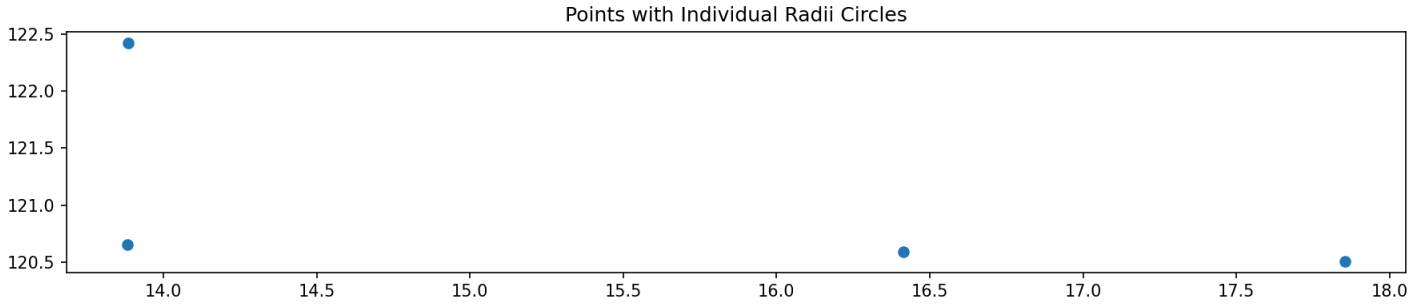
Point	Radius (deg)	Radius (km)
P1	12.814180	1,366.434
P2	12.862389	1,383.889
P3	14.618559	1,572.558
P4	12.720785	1,359.866

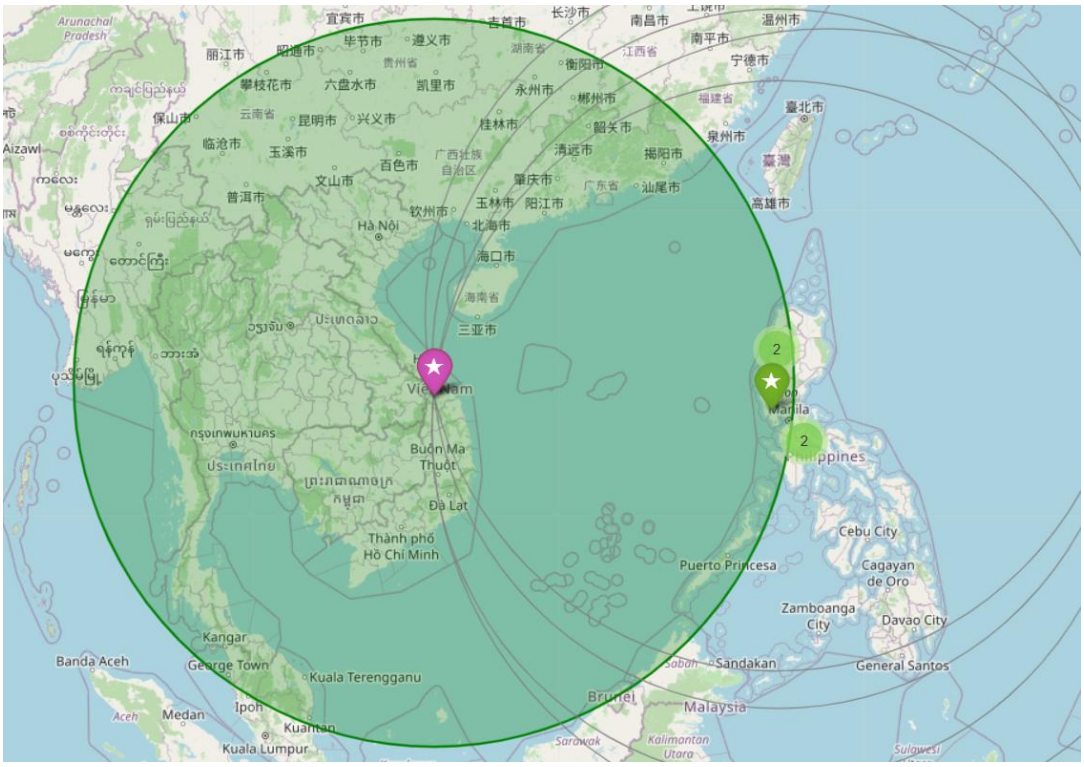
Minimum R: 12.720785° (~1,359.866 km)

Maximum R: 14.618559° (~1,572.558 km)


Original Points
P3
P2
P4
P1
122.5
122.0
121.5
121.0
120.5
14.0
14.5
15.0
15.5
16.0
16.5
17.0
17.5
18.0


Points with Epicenters and Approximate Radius Circles
Approximate Epicenter (H, K)
Measured Epicenter (Hmeasured, Kmeasured)
Recalculated Epicenter (H_r, K_r)
122.5
120.0
117.5
115.0
112.5
110.0
107.5
14.0
14.5
15.0
15.5
16.0
16.5
17.0
17.5
18.0


Points with Individual Radii Circles
122.5
122.0
121.5
121.0
120.5
14.0
14.5
15.0
15.5
16.0
16.5
17.0
17.5
18.0


Hà Nội
Manila
Buôn Ma Thuột
Đà Lạt
Thành phố Hồ Chí Minh
Cebu City
Puerto Princesa
Cagayan de Oro
Zamboanga City
Davao City
General Santos
Kangar
Banda Aceh
George Town
Kuala Terengganu
Ipoh
Medan
Kuantan
Kuala Lumpur
Brunei
Malaysia
Sandakan
2
2

CONSOLIDATED TABLE				
Intensity	Initial Epicenter	Recalculated Epicenter	Outliers	Distance Error
Actual Location	15.02° N, 120.34° E	-	-	-
VIII	15.1544° N, 120.2751° E	15.1544° N, 120.2751° E	None	16.490 km
VII	14.9295° N, 120.3884° E	14.8946° N, 120.4204° E	Only Guagua, Pampanga, and Sasmuan, Pampanga, seem stable	16.402 km
VI	14.5783° N, 120.0659° E	14.5783° N, 120.0659° E	None	57.279 km
V	14.8588° N, 120.8045° E	-	-	53.025 km
IV	15.2399° N, 121.7855° E	15.1305° N, 120.7855° E	A total of 10 outliers: Meycauyan, Plaridel, and San Jose Del Monte, Bulacan; Rosales, Pangasinan; La Trinidad, Benguet; Kasibu Nueva Vizcaya; Bacoor, Imus, Maragondon, Cavite; Antipolo City; Baguio City, Las Pinas City, and Tagaytay City	49.391 km
III	14.6260° N, 120.8618° E	14.7105° N, 121.1852° E	Only Calamba City, Laguna, seems stable	96.833 km
II	14.1374° N, 121.2818° E	14.6718° N, 121.1949° E	Only Dolores Quezon seems stable	99.709 km
I	15.5460° N, 107.9002° E	15.5460° N, 107.9002° E	None	1,335.420 km

PHYSICAL INTERPRETATION BY INTENSITY LEVEL

(Central Luzon Ms 6.1 — April 22, 2019)

The dataset for the Central Luzon earthquake is **large, geographically diverse, and spans multiple provinces**, which makes it an excellent event for studying how intensities behave over complex geological settings (Central Luzon Basin, Zambales Range, Pampanga River Delta, Manila Basin).

INTENSITY VIII — Near-Source Severe Damage Zone

Recalculated Epicenter: 15.1544° N, 120.2751° E
Distance from actual epicenter: ~16.49 km
Localities: Nabuclod, Floridablanca, Diaz, Porac (all Pampanga)

Interpretation

These communities are located on the **western margin of Pampanga**, near the foot of the Zambales Mountains, which correlates directly with the **near-field rupture zone** of the 2019 event.

Porac and Floridablanca—the most brutal hit—lie:

- close to the **Lubao Fault segment**,
- at the **interface of steep topography and deep sedimentary basins**,
- in areas of high site amplification.

Thus, the intense shaking is physically expected.

The recalculated epicenter (16 km error) still falls **close to the Zambales–Pampanga boundary**, indicating that the high intensities retain a physical proximity to the fault rupture.

Key point:
Intensity VIII gives a **tight and physically consistent** constraint on the epicentral region.

INTENSITY VII — Strong Destructive Shaking Belt

Recalculated Epicenter: 14.8946° N, 120.4204° E
Distance Error: 16.402 km
Localities: Porac, Floridablanca, Guagua, Lubao, Sasmuan (Pampanga); San Marcelino, San Narciso (Zambales)

Interpretation

This intensity level spreads into two geological domains:

1. **Pampanga River Delta** (Guagua, Lubao, Sasmuan)
    - Soft sediments

    - Known basin amplification
    - Strong shaking even farther from the fault
2. **Zambales Ophiolite Belt** (San Marcelino, San Narciso)
    - Hard bedrock
    - Efficient transmission of high-frequency waves
    - Directivity effects from the rupture

The recalculated epicenter remains within ~16 km of the actual location—similar to Intensity VIII—indicating that near-source destructive shaking retains **good but slightly weakened geometric fidelity**.

Physical pattern:
Intensity VII outlines the **oval-shaped damage belt** that wraps around the fault at varying distances depending on soil conditions.

INTENSITY VI — Very Strong Regional Shaking

Recalculated Epicenter: 14.5783° N, 120.0659° E
Distance Error: 57.279 km
Localities: Olongapo, Subic, Angeles City, Bacolor, Masantol

Interpretation

Intensity VI towns are now spread widely across Central Luzon:

- **Zambales Coast:** Olongapo and Subic
  → Bedrock, topographic focusing
- **Angeles & Bacolor:** On thick volcanic/depositional sediments
  → High basin response
- **Masantol:** On deep deltaic muds
  → Strong amplification far from the source

This geographic spread introduces *anisotropy* (non-circular propagation), which pushes the inferred epicenter **southwest**, away from the actual source.

The ~57 km deviation is natural at this level because **VI is influenced strongly by local geology**, not just distance to the fault.

Physical takeaway:
Intensity VI marks the beginning of **wide regional shaking**, where soil effects distort the macroseismic geometry.

INTENSITY V — Strong Felt Shaking (Manila Basin + Central Luzon Plains)

Recalculated Epicenter: *No stable epicenter formed*
Distance Error: 53.025 km

Interpretation

Intensity V covers an enormous region:

- Zambales
- Pampanga
- Bataan
- Bulacan
- Cavite
- NCR (Makati, Manila, Mandaluyong, QC, etc.)
- Tarlac
- Lipa City (Batangas)

This band includes:

- **mountain blocks** (Zambales)
- **sedimentary basins** (Pampanga, Manila)
- **coastal plains**
- **volcanic terrains** (Batangas, Tarlac)
- **reclaimed land** (Pasay, Manila)
- **deep basin edges** (QC, Caloocan)

Such heterogeneous geology makes intensity V **physically inconsistent across space**.

Some areas with soft soil (Manila, Calumpit, Minalin) report higher intensities despite being *farther* from the epicenter compared to bedrock areas closer to the source.

Result:
The inversion becomes **unstable**, and no single epicenter satisfies the radii.

Physical cause:
Intensity V reflects **both distance and significant site amplification**, blurring epicentral constraints.

INTENSITY IV — Moderate Regional Shaking (Luzon-wide)

Recalculated Epicenter: 15.1305° N, 120.7855° E
Distance Error: ~49.391 km
Localities: Spread across Bulacan, Pangasinan, Benguet, Nueva Vizcaya, Rizal, Cavite, Batangas, NCR

Interpretation

Intensity IV marks where shaking transitions to **Luzon-wide perceptibility**.

This band includes:

- **mountain provinces** (Baguio, Itogon, La Trinidad)
- **the eastern flanks** (Rodriguez, Antipolo)
- **southern Luzon** (Nasugbu, Tagaytay)
- **central Luzon plains**
- **Manila Basin** (Caloocan, Las Piñas, Marikina)

The variation in intensity IV at great distances is driven by:

- long-period wave trapping in basins,
- topographic amplification in mountains,
- sedimentary thickness variations.

Thus, the inversion identifies a **region** but cannot localize a point.

Physical pattern:
Intensity IV outlines the **outer strong-motion envelope** of the earthquake.

INTENSITY III — Weak Regional Shaking

Recalculated Epicenter: 14.7105° N, 121.1852° E
Distance Error: ~96.833 km

Interpretation

Intensity III towns are located in:

- Quirino
- Nueva Ecija
- Quezon
- Laguna
- Batangas

- Cavite
- Bataan
- Muntinlupa area

This intensity band represents **weak shaking** dominated by:

- human perception
- building resonance
- local soils
- timing of the earthquake

Because these factors overshadow direct distance, the calculated epicenter drifts toward the **center of gravity of reports**, landing near Rizal–Quezon City.

INTENSITY II — Slight Shaking, Far-Field

Recalculated Epicenter: 14.6718° N, 121.1949° E
Distance Error: ~99.709 km

Interpretation

Intensity II observes tiny accelerations, often perceived only in quiet conditions.

These reports come from:

- Aurora
- Quezon
- Camarines Norte
- Pangasinan
- Laguna
- Lucena
  ...spanning >250 km.

This reveals only **felt reach**, not source information.
The inversion drifts toward **southeastern Luzon** due to the sheer number of reports in that area.

INTENSITY I — Barely Perceptible

Recalculated Epicenter: 15.5460° N, 107.9002° E
Distance Error: ~1,335 km

Interpretation

Because Intensity I is influenced by pure perceptibility and non-seismic factors (such as wind, building sway, and human activity), it yields **no physical information** about the earthquake's actual location.

The massive distance error is expected.

CONSOLIDATED PHYSICAL INTERPRETATION

The Central Luzon earthquake provides a textbook example of how macroseismic fields behave in a **large, geologically diverse region**.

1. High intensities (VIII–VII) give consistent and physically meaningful epicentral constraints.

Their recalculated epicenters are ~16 km from the proper location.
These towns lie close to the **Lubao–Porac fault system**, capturing near-field rupture characteristics.

2. Intensity VI marks the transition where strong shaking is still tied to the source but influenced heavily by soil conditions.

The error increases to ~57 km because this intensity extends into the **Pampanga and Manila basins**, where shaking no longer correlates purely with distance.

3. Intensities V–IV cover multiple provinces and geological domains.

These levels mix:

- mountain ranges
- sedimentary basins
- reclaimed lands
- volcanic terrains

Because of their broad distribution and heterogeneity, the inversion spreads away from the actual source.

4. Intensities III–II reflect regional perceptibility, not source proximity.

At these distances, intensities no longer follow attenuation laws.
Errors grow to ~100 km.

5. Intensity I reports cannot be used for epicenter estimation.

They reflect only the outermost perceptibility fringe.

Overall Physical Conclusion

The intensity field of the 2019 Central Luzon earthquake shows a **clear, physically logical attenuation pattern**:

- **VIII–VII:** tightly constrain the epicenter
- **VI:** still directional but distorted by basin effects
- **V–IV:** widespread, heterogeneous shaking
- **III–II:** regional felt reports without geometric value
- **I:** not seismologically useful